Report R16 — A Plan Written Blind, Then Executed

ACRIN 6668 sealed holdout. An isolated agent wrote the analysis from schema alone; the plan was hashed before any outcome was read; the verdict is CONFIRMED.

16 August 2026 • Quantara

Summary

This is the second cohort for the autonomous-generation arm, and the first run of a *structurally* separated generator. An agent whose entire readable surface was one JSON file — form names, column names, row counts and the trial's data dictionary, with no cell value from any outcome field — wrote a 493-line pre-registered analysis plan for a cohort it had never seen. The plan was SHA-256 hashed at `752b7462fce56899...cb74e6770` on 15 August at 13:07:14 Z, before the coordinator read a single outcome value. It was then executed as written.

Verdict: CONFIRMED. Higher post-treatment FDG peak SUV predicts shorter overall survival: **HR 1.202 per doubling** (95% CI 1.100–1.315), permutation $p = 0.0005$, on 166 analysed patients with 125 events. That is the association ACRIN 6668 was designed to test, recovered by a plan written in ignorance of the data's contents.

All nine integrity gates passed. Three of four positive controls corroborated, the fourth declared uninformative by its own pre-registered rule rather than quietly dropped. All seven pre-declared robustness tests agree in direction and survive Holm correction.

1 What the generator produced, and what it caught

The plan is in `evidence/PLAN_hashed_752b7462.yaml`. From schema and dictionary alone it identified twelve traps in advance. Four mattered:

- **Immortal time.** The exposure is measured 8–20 weeks after chemoradiotherapy, so measuring survival from registration would guarantee survival to exposure. The plan set the time origin at the post-treatment PET (a landmark) and made $\min(T) \geq 0$ a gate.
- **Vital-status code 9** means "dead, date of death unknown". The obvious `event = (f1e2 == 2)` silently undercounts deaths. The plan treated 9 as an event and made a censoring sensitivity a *condition* of confirming.
- **Cutoff forking.** Form SS carries 1,185 rows keyed by an SUV inclusion cutoff — five analyses per patient, a ready-made garden of forking paths. The plan locked SS to the single most inclusive cutoff and forbade any comparison across cutoffs.
- **Undocumented uniqueness.** The recon package did not state that form LE is one row per patient, yet the primary exposure depended on it. The plan required verification and pre-specified a duplicate-resolution rule.

That last one fired: LE has 244 rows over 243 patients. The plan's rule resolved it and reported the single affected patient.

2 Results

2.1 Gates

Gate	Check	Result
G1	two release samples recombine, disjoint, A0 = 251 rows / 251 patients	PASS
G2	six join counts reproduced exactly; 0 orphans in any form	PASS
G2b	LE duplicates resolved by the declared rule (1 patient)	PASS
G3	SS confirmed keyed by cutoff (values 0/2/2.5/3/3.5; modal 5 rows per patient)	PASS
G4	no outcome-period leakage; exposure hash unchanged from the freeze	PASS
G5	$251 = 166 + 85$, ledger reconciles exactly	PASS
G6	permutation null centred (mean z 0.034, SD 0.982, rejection 0.042)	PASS
G7	exposure coverage ≥ 180 (observed exactly 180)	PASS
G8	vital-status tally: 125 dead, 38 alive, 3 lost; median follow-up 49.2 months	PASS
G9	primary point estimate lies inside its own CI *(added after a bug, §5)*	PASS

2.2 Exclusion ledger

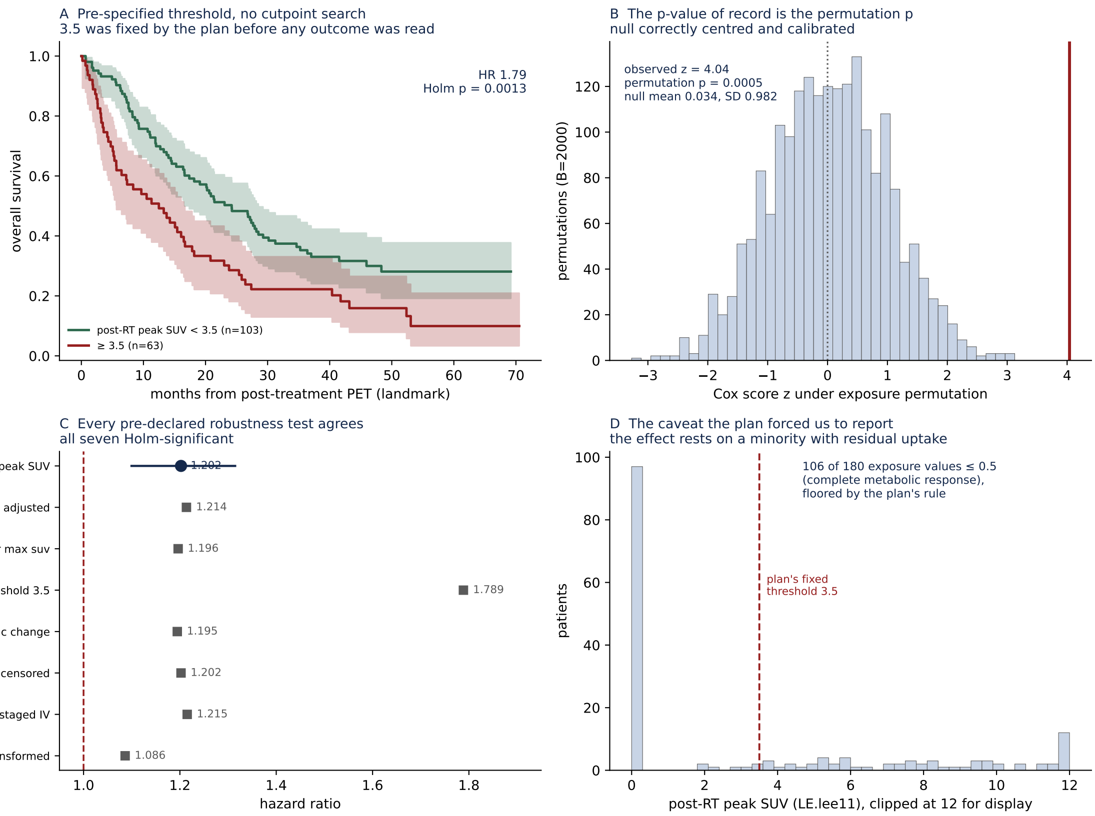
E2	ineligible / duplicate registration	18
E3	no radiotherapy commenced	15
E4	no post-treatment PET timepoint (time origin undefined)	38
E5	post-RT peak SUV missing	10
E8	PET outside the 28–280 day post-RT window	4
	E1, E6, E7	0
Total excluded		**85**
Analysed		**166**

2.3 Primary and controls

	Test	HR	Outcome
Primary	OS on $\log_2$ post-RT peak SUV	**1.202**	perm. $p = 0.0005$ 95% CI 1.100–1.315
PC0	post-RT SUV below pre-RT (outcome-free)	—	**PASS** (94.4%)
PC1	progression by day 365 → shorter OS	2.463	**PASS** ($p = 3.8 \times 10^{-5}$)
PC2	performance status ≥ 1 vs 0	1.382	PASS-WEAK ($p = 0.072$)
PC3	weight loss $\geq 5\%$	—	UNINFORMATIVE-BY-N (26 exposed)

PC1 is the one that matters: it validates that the survival clock, the per-patient reduction over 6.3 follow-up rows per person, and the vital-status map were assembled correctly. Its near-tautological clinical content is the point — it fails if the plumbing is wrong.

PC2 came out in the right direction but short of significance, and PC3 was declared uninformative because only 26 patients had $\geq 5\%$ weight loss against a pre-declared threshold of 30. The plan had *anticipated* both: trial eligibility screens performance status, so range restriction was expected and given its own outcome branch in advance rather than being explained afterwards.



A Survival split at the plan's fixed 3.5 threshold — no cutpoint search was permitted. **B** The permutation null and the observed statistic. **C** Primary and all seven robustness tests. **D** The exposure distribution, and the caveat the plan forced into the open.

2.4 Robustness

All seven pre-declared secondary tests, Holm-corrected across the family:

	Test	**HR**	**Holm** p	n
S1	adjusted (age, sex, stage, performance status)	1.214	0.0002	165
S2	tumour max SUV instead of peak	1.196	0.0002	166
S3	fixed threshold 3.5 (no search)	1.789	0.0013	166
S4	metabolic change, post minus pre on $\log_2$	1.195	0.0002	165
S5	code-9 deaths censored instead of events	1.202	0.0002	166
S6	excluding patients upstaged to IV	1.215	0.0002	161
S7	untransformed SUV (per unit)	1.086	< 0.0001	166

The adjusted estimate is slightly *larger* than the unadjusted one, and the effect survives excluding upstaged patients, changing the SUV field, and changing the transformation.

3 The caveat the plan forced us to report

106 of 180 exposure values were at or below 0.5 — complete metabolic response — and were floored by the plan's rule. So 59% of the cohort sits at a single tied value and the association is carried by the minority with residual uptake. The plan required this count to be reported, which is why it appears here rather than in a reviewer's letter. S3, the binary split at a fixed 3.5, is arguably the more interpretable form of the same finding: median survival is clearly separated (Figure A) with HR 1.79.

4 Two coordinator errors, and one that gates did not catch

Recorded because this report is partly about process, and the process failed in ways worth knowing.

Two implementation errors were mine, not the plan's. First, I wrote LE uniqueness as a hard stop when the plan had already anticipated the duplicate and specified a remedy — the plan had thought further ahead than my executor. Second, the landmark date came back empty because the CSVs capitalise day-offset fields (`TAe7d`) differently from the dictionary the plan quotes (`tae7d`); same field, different case. Neither required reinterpreting the plan.

One error the gates did not catch. An earlier execution reported the primary as HR 1.382 with CI 1.050–1.124 — a point estimate outside its own interval. Inside the control blocks I had reused `hr` and `ci` as scratch names, clobbering the primary's values before the verdict logic read them, so the verdict was briefly evaluated using PC2's hazard ratio. All eight of the plan's gates passed while this was true. The conclusion survives correction, but it was momentarily reached by accident.

What caught it was not a gate but an internally impossible number. Gate G9 was added afterwards to assert that the primary point estimate lies inside its own confidence interval. **A pre-registration protects against choosing your analysis after seeing results; it does not protect against implementing it wrongly.** Those are different failure modes and need different defences.

5 Limitations

Single trial, one modality, tabular only. This used the core-lab PET assessments, not the 135.5 GiB of PET/CT imaging. The generalisation claim concerns an unseen cohort and data shape, not an unseen imaging pipeline.

The exposure is 59% floored (§4).

Two of four controls did not fully corroborate. PC2 weak, PC3 uninformative. Under the plan's aggregate rule this is sufficient for CONFIRMED, because PC2 was in the right direction and PC1 passed strongly — but a fully corroborated run would have been stronger.

G7 passed at exactly its threshold (180 against ≥ 180). One patient fewer and the declared fallback chain would have fired, changing the exposure source.

Not an independent replication of the trial. The published ACRIN 6668 analysis used the same data. The claim here is about the *process* — that a blind, pre-registered plan recovers a known result — not about establishing the biology afresh.

Reproducing this report

`evidence/PLAN_hashed_752b7462.yaml` (the plan as hashed), `evidence/PREREGISTRATION.json` (hash and timestamp), `evidence/execute_plan.py` (the executor), `evidence/exposure_table.csv` (frozen at Stage 1, SHA-256 `854378633ea6992c...`), `evidence/analysis_frame.csv`, `evidence/permutation_` (all 2,000 null statistics), `evidence/gates.json`, `evidence/results.json`. Figure: `python3 make_figure.py`.